BUSINESS REQUIREMENTS DOCUMENT

ERP Sample Project

Feature Summaries
Prepared by Kolosal AI

Version 1.0 · July 27, 2026 · Draft · Internal

BUSINESS REQUIREMENTS DOCUMENT

Product List Page

Prepared by Kolosal AI

Version 1.0 **Date** July 27, 2026 **Status** Draft **Classification** Internal

1. Frontend

1.1 Overview

The Product List Page is the main screen where users can browse, search, sort, and page through all products in the catalog. It brings together a navigation menu, a set of search filters, a results table, and page-turning controls so users can quickly find the products they're looking for.

1.2 Step-by-Step Flow

1. When the page opens, it automatically loads the first 20 products, sorted by newest first, and displays them in a table.
2. The page shows two navigation menus (Main Menu and Sub Menu) with the "Products" section highlighted so users know where they are.
3. An "Add Product" button in the top corner lets users jump straight to a page for creating a new product.
4. Above the table, a filter bar lets users type in a SKU, product name, or product code, and choose a brand or category. As they type, the system checks their entries are valid (right length, allowed characters) and shows an error message if not, disabling the search button until it's fixed.
5. Some filter options (Product Type, Tax, Tags) are visible on the filter bar for visual consistency but don't yet actually filter results, since the search system doesn't support them yet.
6. When the user clicks "Apply Search," the system combines the SKU, name, and product code entries into one search request, includes any selected brand/category, and fetches a fresh, updated list of products—resetting back to the first page of results.
7. The results table displays each product's SKU, name, brand, product code, tax, price, status, and stock quantity. The SKU and name are clickable links that take the user to that product's detail page.
8. Users can click column headers (like Name, Status, Price, Stock) to sort the list. The first click sorts smallest-to-largest (or A-to-Z), and clicking the same column again reverses the order.
9. If there are more products than fit on one page, pagination controls let users move to the next or previous page, automatically disabling buttons when there is no next or previous page available.
10. If something goes wrong while fetching data (e.g., a server error), an error message appears above the table, but any previously loaded products remain visible so the user isn't left with a blank screen.
11. While data is loading, a loading indicator is shown; if a search returns no results, the table displays a "No products found" message.

1.3 Key Concepts

- Filter Bar: The area above the table where users enter search criteria (SKU, name, product code, brand, category) before running a search.
- Apply Search: The action that takes whatever the user has typed/selected and fetches a matching list of products, restarting from page one.
- Sorting: Clicking a column heading reorders the table by that column, alternating between ascending and descending order.
- Pagination: Breaking a large product list into pages (20 items at a time) with "Previous/Next" navigation.
- Orphan Fields: Filter options (Product Type, Tax, Tags, Price Range) that appear on screen for design consistency but are not yet connected to actual search functionality—flagged for future development.
- Error Handling: If a search or page load fails, users see a clear error message while the system keeps showing the last successful set of results instead of clearing the screen.

1.4 Business Value

This feature gives staff a fast, reliable way to find and manage products in a large catalog without needing technical knowledge. By combining search, filtering, sorting, and pagination in one place, it reduces the time spent hunting for specific items, minimizes errors through built-in input validation, and ensures the system remains usable even when errors occur. It also creates a clear, consistent entry point for adding new products, supporting smooth day-to-day catalog management.

2. Logic

2.1 Overview

This feature powers the Product List Page, allowing users to browse, search, and filter all products in the system. When the page first loads, it automatically retrieves a list of products. Users can also refine that list at any time by entering search terms or applying filters (like category, status, or brand) and clicking "Apply Search." Both actions use the same underlying process to fetch and display product information, ensuring consistent behavior throughout.

2.2 Step-by-Step Flow

1. The page loads, or the user clicks "Apply Search" after entering filter criteria.
2. The system reads whatever search terms, filters, sorting preferences, and page number the user has specified (or uses sensible defaults if none are given, such as showing page 1 with 20 products, sorted by newest first).
3.

The system checks that the requested page number and page size make sense (e.g., not zero or negative). If not, it shows an error message.

4. The system searches the product records, applying:

- Any free-text search across product name, SKU, brand, and product code.
- Any specific filters selected (category, status, brand).
- The chosen sort order.
- The requested page of results.

5. The system counts how many total products match the search/filter criteria (regardless of pagination), so the user knows how many results exist in total.
6. The matching products are formatted into a consistent structure for display, showing details like SKU, name, brand, product code, tax, price, category, stock quantity, and status.
7. The formatted list, along with the total count and current page details, is sent back to the screen for the user to view.
8. If something goes wrong during this process, the system logs the issue internally and shows a generic error message to the user, without exposing technical details.

2.3 Key Concepts

- Search: A free-text lookup that scans product name, SKU, brand, and product code to find matches, making it easy to find products without knowing exact details.
- Filters: Specific criteria (category, status, brand) that narrow down the product list to only those matching selected values.
- Pagination: Splitting a large list of products into smaller, manageable pages (e.g., 20 products at a time) so the page loads quickly and is easier to navigate.
- Sorting: The ability to order the product list by a chosen field (like creation date), either ascending or descending.
- Mapping Rules: The defined way each piece of product data (like SKU, name, brand, price) is matched to its corresponding label on the screen, ensuring the right information appears in the right place.
- Total Count: The full number of products that match the current search/filter, separate from how many are shown on the current page, so users understand the complete scope of results.

2.4 Business Value

This feature gives business users fast, reliable access to product information without needing technical knowledge or direct database access. By supporting both initial browsing and refined searches through a single, consistent

process, it reduces development complexity, minimizes bugs, and ensures a smooth, predictable user experience. Employees can quickly locate specific products among potentially thousands of items, improving operational efficiency in inventory management, sales support, and reporting.

3. Data

3.1 Overview

This feature powers the Product List Page, where users can browse, search, and filter through all products in the system. It defines how product information is stored, how the system finds and organizes that information when requested, and how results are delivered back to the screen in a fast, organized, and predictable way — complete with pagination so large product catalogs remain easy to navigate.

3.2 Step-by-Step Flow

1. Product information (such as SKU, name, brand, product code, tax, price, category, stock level, and status) is stored in a structured, organized way so it can be reliably retrieved later.
2. When a user visits the Product List Page, the system prepares to fetch a list of products based on whatever search terms or filters the user has selected.
3. If the user has typed a search term, the system looks for matches in the product name, SKU, or product code.
4. If the user has selected filters (such as brand, category, status, or a price range), the system narrows the results to only include products matching those criteria.
5. The system counts how many total products match the search and filters, so it knows how many pages of results exist.
6. The system then retrieves only the specific "page" of results the user is currently viewing (for example, results 21–40 if the user is on page 2), sorted alphabetically by product name.
7. The system also provides supporting information to populate filter dropdown menus — specifically, the full list of available brands and categories — so users can easily pick valid filter options.
8. The final list of products, along with the total count and pagination details, is sent to the screen for display, following a defined mapping so each piece of data lands in the correct column (e.g., SKU shows under "SKU," productName shows under "Name," and so on).
9. Users can also retrieve a single product's full details on demand (e.g., when clicking into a product), separate from the list view.

3.3 Key Concepts

–

Filters: Criteria users can apply to narrow down the product list, such as brand, category, status, or a price range.

- Search: A free-text lookup that checks the product name, SKU, and product code to find relevant matches.
- Pagination: Breaking a large list of products into smaller, manageable "pages" so the page loads quickly and is easy to browse.
- Total Count: The number of products that match the current search/filter criteria, used to determine how many pages of results exist.
- Filter Dropdowns (Brand/Category lists): Pre-populated lists of all available brand names and categories, so users can select valid filter values rather than guessing.
- Data Mapping: A defined rule that ensures each piece of product data (like price or SKU) always appears in the correct, expected place on the screen.

3.4 Business Value

This feature ensures that anyone browsing the product catalog — whether managing inventory, checking pricing, or searching for a specific item — can quickly find what they need without wading through irrelevant results. By supporting fast search, meaningful filters, and manageable page sizes, it improves efficiency for staff who rely on accurate, organized product data daily. It also lays a consistent foundation (clear field names and structured queries) that reduces errors and rework as the system grows or as new filters and reports are added in the future.

BUSINESS REQUIREMENTS DOCUMENT

Products Menu

Prepared by Kolosal AI

Version 1.0 **Date** July 27, 2026 **Status** Draft **Classification** Internal

1. Frontend

1.1 Overview

This feature covers the top navigation menu in the ERP system's header — specifically the row of clickable items like "Products" and its neighboring menu options. It controls what happens when a user clicks on any of these navigation items: which item gets highlighted, which page content appears, and whether a dropdown-style sub-menu opens underneath. This is purely a visual, on-screen behavior — no information is being saved or retrieved from any database or server; everything happens instantly on the user's screen.

1.2 Step-by-Step Flow

1. When the page first loads, the "Products" menu item is automatically shown as the selected (active) item, and because "Products" has a sub-menu, that sub-menu is already open, displaying its default page right away — no click required.
2. The navigation bar displays each menu item in order (label and icon), following the intended visual design.
3. If an item is currently the active/selected one, it's shown with a highlighted style so users can easily tell where they are.
4. If an item has a sub-menu attached to it, a small arrow (chevron) appears next to its label. This arrow points down when the sub-menu is closed and flips upward when it's open.
5. When a user clicks any navigation item, the system immediately marks it as the active item and updates the main content area to show that item's default page.
6. If the clicked item has a sub-menu:

– Clicking it again while already open will close the sub-menu.

– Clicking it while closed will open it, showing its default sub-page.

– Opening a sub-menu automatically closes any other sub-menu that was open, since only one can be open at a time.

7. If the clicked item does not have a sub-menu, the system immediately closes any sub-menu that might be open elsewhere, since the user's focus has moved elsewhere.
8. All these visual updates (highlighting, arrow direction, sub-menu opening/closing, page switching) happen instantly with no waiting, spinners, or delays, since nothing is being fetched from outside the application.

1.3 Key Concepts

– Navigation Item: A clickable entry in the header menu (e.g., "Products") that represents a section of the application.

– Active State: A visual indicator (like a highlight) showing which navigation item the user currently has selected.

–

Sub-Menu (Accordion Behavior): A collapsible list of additional options tied to a navigation item, which expands or collapses like an accordion — only one sub-menu can be open at any given time.

- Chevron Indicator: The small arrow icon that tells users whether a menu item's sub-menu is open (pointing up) or closed (pointing down).
- Default Route/Page: The page automatically shown when a navigation item (or its sub-menu) is selected, ensuring users always land somewhere meaningful.
- Client-Side State: All the behavior described happens instantly within the user's browser/app session — there's no backend data being fetched or saved for this feature.

1.4 Business Value

This navigation menu gives users a fast, intuitive, and responsive way to move between different sections of the ERP system's header, particularly the Products area and its related pages. Because everything updates instantly without loading delays, the experience feels smooth and professional. Setting "Products" as the default active view also ensures users immediately see relevant content when they open the app, reducing extra clicks and improving overall usability. Since this menu is self-contained and doesn't rely on external data, it's also simpler to maintain, test, and update as the product evolves.

BUSINESS REQUIREMENTS DOCUMENT

Main Menu

Prepared by Kolosal AI

Version 1.0 **Date** July 27, 2026 **Status** Draft **Classification** Internal

1. Frontend

1.1 Overview

This feature describes the Main Menu — the header bar that appears at the top of every page in the ERP system. It gives users a consistent way to navigate between sections, search for information, check notifications, and manage their account, no matter which page they're on.

1.2 Step-by-Step Flow

1. Viewing the header: Every page displays the same top bar, containing the company logo, navigation links, a search box, a notification bell, and an account avatar.
2. Clicking "Products": When a user clicks the Products navigation item, a menu panel opens beneath it showing related options. Clicking it again closes the panel.
3. Clicking a navigation item: When a user clicks any other top navigation item (like a section link), the system marks that item as "active" (highlighted), closes any open panels, and takes the user to the corresponding page.
4. Using the search box: As a user types into the search field, the system trims extra spaces and limits entry to 200 characters. It waits briefly after the user stops typing (to avoid reacting to every keystroke) before treating the typed text as a search request. If the box is cleared, any search results are cleared too.
5. Opening notifications: Clicking the bell icon opens a dropdown meant to show unread alerts, and closes any other open panels. Clicking again closes it.
6. Opening the account menu: Clicking the avatar opens a dropdown with "Settings" and "Sign Out" options, closing any other open panels. Clicking again closes it.
7. Only one panel open at a time: Whenever the user opens a Products menu, notification panel, or account menu, any other currently open panel automatically closes — keeping the header uncluttered.

1.3 Key Concepts

- Active navigation state: The system remembers which navigation item was last clicked so it can visually highlight it, helping users know where they are.
- Debounced search: Instead of searching on every keystroke, the system waits for a short pause in typing before acting — this prevents unnecessary or premature searches while typing.
- Mutually exclusive dropdowns: Only one dropdown (Products, Notifications, or Account) can be open at a time; opening one automatically closes the others.
- Placeholder features: The notification list and account profile details (like name and photo) are designed to appear in the header, but the actual data connections for these are not yet defined — they will be completed once that information becomes available.

- Input safeguards: The search box automatically trims stray spaces and caps entries at 200 characters to keep input clean and manageable.

1.4 Business Value

This header creates a consistent, predictable navigation experience across the entire ERP system, so employees always know where key actions (search, notifications, account, navigation) are located, regardless of which module they're using. The debounced search reduces unnecessary system load, keeping the interface responsive. Auto-closing of dropdown menus keeps the interface clean and prevents visual clutter or user confusion. Although notifications and account details aren't fully wired to live data yet, the structure is ready to support these once the underlying data becomes available — minimizing future rework.

BUSINESS REQUIREMENTS DOCUMENT

Products Sub Menu

Prepared by Kolosal AI

Version 1.0 **Date** July 27, 2026 **Status** Draft **Classification** Internal

1. Frontend

1.1 Overview

This feature is a small navigation menu that appears in the header of the ERP system, under the "Products" section. It works like a dropdown or accordion list that lets users switch between two views: a general "Products" list and a "Service Products" list. Everything happens instantly on the user's screen — there's no waiting, no server requests, and no data entry involved. It's purely about helping users move between two related pages quickly and clearly.

1.2 Step-by-Step Flow

1. The user sees a "Products Menu" heading in the header, which acts as an expandable/collapsible section (an accordion).
2. When the user clicks on this heading, the section opens or closes, revealing (or hiding) two navigation options: "Products" and "Service Products."
3. By default, when the page first loads, the menu is already open and "Products" is automatically selected and highlighted — so the Products List Page is shown right away without the user needing to click anything.
4. If the user clicks "Products," the system highlights that option and displays the Products List Page.
5. If the user clicks "Service Products," the system highlights that option instead and displays the Service Products List Page.
6. Whichever option is selected, the visual highlight and the page shown update immediately and stay in sync with each other — clicking a menu item always shows the matching page right away.
7. Users can also navigate this menu using a keyboard (not just a mouse), ensuring accessibility for all users.

1.3 Key Concepts

- Accordion menu: A collapsible section that expands to show more options and collapses to hide them, used here for the "Products Menu."
- Active state: A visual indicator (like a highlight or bold text) showing which menu option is currently selected.
- Default selection: The "Products" option is automatically chosen when the page loads, so users immediately see the Products List Page without extra clicks.
- Synchronized navigation: The menu selection and the displayed page are always kept in sync — selecting an item and seeing its matching page happen together, instantly.
- Client-side behavior: All actions (opening the menu, switching pages) happen directly on the user's screen with no delay, since there's no data being fetched from a server for this menu itself.

1.4 Business Value

This feature gives users a fast, intuitive way to move between related product views without confusion or delay. By automatically showing the most relevant page (Products) by default, it reduces extra clicks and helps users get to what they need faster. Because the menu responds instantly and works well with keyboards, it also improves accessibility and overall user experience, reflecting well on the professionalism and usability of the ERP system's interface.

BUSINESS REQUIREMENTS DOCUMENT

GET /api/products

Prepared by Kolosal AI

Version 1.0 **Date** July 27, 2026 **Status** Draft **Classification** Internal

1. Logic

1.1 Overview

This feature allows a user or system to retrieve a list of products stored in the ERP system. Instead of returning every product at once (which could be slow or overwhelming), the system returns products in smaller, manageable "pages" and includes helpful information about how many products exist in total. It also includes checks to make sure only authorized users can access the data and that any requests are properly formatted.

1.2 Step-by-Step Flow

1. The system first checks whether the person or system making the request is properly logged in and authorized. If not, it immediately stops and returns an "Unauthorized" message.
2. The system looks at the request to see if the user specified which "page" of results they want and how many products ("limit") should appear per page.
3. If the user didn't provide these details, the system automatically defaults to page 1 and 20 products per page.
4. The system checks that the page and limit values are valid positive numbers. If not, it returns an error explaining that the request was malformed.
5. To prevent overly large requests, if someone asks for more than 100 products per page, the system automatically limits it to 100.
6. The system then retrieves the requested page of products from the database, ordered so the most recently added products appear first.
7. If the requested page has no products (for example, asking for page 10 when there are only 2 pages), the system returns a "Not Found" message.
8. If everything is valid, the system builds a response containing the list of products (with details like name, price, stock, category, etc.) along with pagination information (current page, limit, total number of products, and total number of pages).
9. This complete response is sent back to the user with a success status.
10. If something unexpected goes wrong during this process, the system logs the issue internally and returns a generic "Server Error" message to the user, without exposing technical details.

1.3 Key Concepts

– Pagination: A way of splitting a large list of products into smaller, easier-to-manage pages rather than returning everything at once.

–

Page and Limit: "Page" tells the system which section of the product list to show, and "limit" tells it how many products to include per page.

- Authentication Check: A security step to confirm that only authorized users can access the product data.
- Default Values: If a user doesn't specify how many products or which page they want, the system automatically uses sensible defaults (page 1, 20 products).
- Upper Limit Safeguard: A rule that prevents someone from requesting an excessively large number of products in one go, which protects system performance.
- Error Handling: The system provides clear, categorized responses when something goes wrong — such as invalid input, unauthorized access, missing data, or unexpected technical issues.
- Response Structure: The returned information always includes both the actual product data and metadata about pagination, so users know exactly what they're looking at and how to navigate further pages.

1.4 Business Value

This feature ensures that anyone accessing the product catalog gets a fast, reliable, and secure experience — even if the store has thousands of products. By breaking large amounts of data into pages, the system stays efficient and responsive. Built-in security checks protect sensitive business data, while clear error messages help developers and integrators quickly identify and fix issues. Overall, this creates a stable foundation for other parts of the ERP system (like inventory management, sales, or reporting) to reliably access up-to-date product information.

2. Data

2.1 Overview

This feature allows the system to retrieve a list of products stored in the database and return it to whoever requested it (such as a website or app screen showing a product catalog). Instead of returning every product at once, the list is delivered in manageable "pages" (like page 1, page 2, etc.), and the system clearly reports back what went right or wrong at each step — whether the request succeeded, was malformed, found nothing, or hit an unexpected technical problem.

2.2 Step-by-Step Flow

1. A request comes in asking for the product list, optionally specifying which "page" of results to view and how many products to show per page.
2. The system checks that the page number and page size are valid (e.g., positive numbers, and not requesting an unreasonably large amount at once). If these are invalid, the request is rejected with a "Bad Request" response.
3.

If the inputs are valid, the system calculates which "slice" of the product list to retrieve based on the requested page and page size.

4. The system then looks up the matching products in the database, along with a total count of how many products exist in total (not just on this page).
5. If something goes wrong while accessing the database (e.g., a connection issue), the system reports an unexpected server error.
6. If the database lookup succeeds but there are zero products in total, the system reports that no products were found.
7. If products are found, the system returns the list of products for the requested page, along with details like the total number of products, the current page, and how many are shown per page.
8. This information is then handed off to be formatted and displayed correctly on the requesting screen (e.g., a product catalog page), ensuring the page numbers and counts match exactly what the user expects to see.

2.3 Key Concepts

- Pagination: A way of splitting a large list of products into smaller, easier-to-load "pages" instead of returning everything at once.
- Page and Limit: "Page" is which segment of results you want (e.g., page 2), and "limit" is how many products should appear per page.
- Total Count: The overall number of products that exist, used to show users how many pages are available.
- Active Product Flag: A marker showing whether a product should currently appear in listings (e.g., discontinued items might be hidden).
- Error Handling by Scenario: The system distinguishes between different problems — invalid input, no data found, or unexpected technical failures — and responds appropriately to each.
- Data Contract: A fixed, agreed-upon structure (product list, total count, page, limit) that ensures the information handed between the backend and the user interface always matches up correctly.

2.4 Business Value

This feature ensures that anyone browsing the product catalog gets fast, reliable, and predictable results — even if there are thousands of products. By using pagination, the system avoids slowing down or overwhelming the user with too much data at once. Clear, consistent error handling (e.g., distinguishing between "no products exist" and "something went wrong") improves troubleshooting and creates a smoother, more trustworthy experience for both business users and customers relying on accurate product information.

BUSINESS REQUIREMENTS DOCUMENT

GET /api/products?utf8= &q[attribute_predicate]=value&...&page_rows=

Prepared by Kolosal AI

Version 1.0 **Date** July 27, 2026 **Status** Draft **Classification** Internal

1. Logic

1.1 Overview

This feature powers the product search function within the ERP system. It allows users to search and filter the product catalog using a variety of criteria — such as SKU, product name, product code, brand, type, tax category, and tags — while also giving them control over sorting and how many results appear per page. The system carefully checks who is asking, makes sure the request makes sense, and only then returns matching products in an organized, paginated list.

1.2 Step-by-Step Flow

1. The system first confirms the person making the request is properly logged in. If not, access is denied.
2. It then checks whether this person is actually allowed to view the product list. If they lack permission, the request is rejected.
3. The system checks whether this user has been making too many requests too quickly, and blocks the request if they've hit a limit.
4. The incoming search filters (like SKU, name, brand, tags, sorting instructions, and page size) are read and checked to make sure they are in the correct format and type — for example, that an ID is actually a number, not text.
5. If any filter is broken or unreadable, the request is rejected with an error explaining the problem.
6. If filters reference specific items — such as a particular brand, product type, tax rate, or tag — the system verifies those items actually exist in the system. If not, a "not found" error is returned.
7. Once everything is validated, the system builds a precise search request combining all the provided filters (for example, "name contains X" and "brand equals Y").
8. If the user specified sorting preferences (like sorting alphabetically by name), that sort order is applied; otherwise the newest products are shown first by default.
9. The system applies pagination settings, determining which page of results to show and how many items to include per page.
10. The system runs the search against the product data, retrieving matching results and the total count of matches.
11. If something unexpected goes wrong while retrieving the data, a generic server error is returned.
12. Finally, the results are formatted into a clean, standardized list of product details along with pagination information, and sent back to the user.

1.3 Key Concepts

- Authentication: Confirms the person making the request is logged in with valid credentials.
- Authorization: Confirms the logged-in person has permission to view product listings.
- Rate Limiting: Prevents any single user from overwhelming the system with too many requests in a short time.
- Filter Validation: Ensures each search filter (like brand ID or tag ID) is in a usable, correctly formatted state before searching.
- Referenced Data Check: Confirms that any specific brand, type, tax, or tag mentioned in a search actually exists before using it as a filter.
- Search Criteria: The combined set of filters used to narrow down which products should be returned.
- Sorting: Determines the order in which matching products are displayed (e.g., alphabetically or by newest).
- Pagination: Breaks up large result sets into manageable pages, showing only a limited number of products at a time.
- Response Payload: The final, organized set of product information sent back to the requester, including total match count and paging details.

1.4 Business Value

This feature gives users a fast, reliable, and secure way to search the product catalog using the criteria that matter most to their work — whether that's finding products by brand, type, tags, or free-text details. By validating inputs and checking permissions upfront, it prevents confusing errors, protects sensitive operations, and ensures the system remains stable even under heavy or malicious use. Ultimately, this leads to a smoother, more trustworthy experience for anyone managing or browsing the product catalog.

2. Data

2.1 Overview

This feature powers the product search functionality in the ERP system, allowing users or connected systems to look up products in the catalog using a variety of filters — such as SKU, name, product code, brand, type, tax category, tags, and custom info fields. It checks that the request is well-formed, that the person making the request is allowed to do so, and that any filters used actually match real data before returning a page of matching products.

2.2 Step-by-Step Flow

1. Check the request format — The system first confirms the incoming search request is structured correctly (no unexpected or malformed parts). If not, it rejects the request immediately.
2.

Validate filter values — Each filter is checked for the right type of value (for example, brand ID must be a number, and tag lists must be a list of numbers). Sorting instructions and the results-per-page setting are also checked to make sure they make sense.

3. Confirm referenced items exist — If someone searches by a brand, product type, tax category, or tag, the system checks that those actually exist in the catalog. If a referenced item doesn't exist, the search is rejected with a clear explanation.
4. Verify the requester's identity and permissions — The system checks that the caller is properly logged in and has permission to view the product list. Unauthorized or unpermitted requests are blocked.
5. Apply rate limiting — The system checks whether the caller has made too many requests recently, to protect the system from overuse or abuse.
6. Search and retrieve products — If all checks pass, the system searches the product catalog using the given filters, removes duplicate results (which can happen when filtering by tags), sorts the results as requested (or by default order), limits the results to a manageable page size, and returns the matching products.
7. Handle unexpected failures — If something goes wrong unexpectedly while retrieving the data, the system logs the issue internally and returns a generic error message instead of exposing technical details.

2.3 Key Concepts

- Filter validation: Ensuring each search criterion (like brand ID or tag list) is in the correct format before searching, preventing bad or confusing data from causing errors.
- Referential check: Confirming that any ID used in a filter (such as a brand or tax ID) actually corresponds to something that exists in the system.
- Authorization gate: A security checkpoint that verifies the requester is logged in and has permission to view product data.
- Rate limiting: A protective measure that limits how many searches a single caller can perform in a short time, to avoid system overload or misuse.
- Pagination (page_rows): Controls how many products are returned at once, making large result sets easier to manage.
- Sorting (s parameter): Lets the caller specify how results should be ordered, such as alphabetically by product name.
- Duplicate removal: Since products can have multiple tags, the search ensures each product appears only once in the results, even if it matches multiple tag filters.

2.4 Business Value

This feature gives users a reliable, flexible, and secure way to search the product catalog. By validating inputs early, it prevents bad data from causing confusing or incorrect results. By checking permissions and rate limits, it protects sensitive product data and system performance from misuse. And by supporting flexible filtering, sorting, and pagination, it helps users quickly find exactly the products they need — improving efficiency and reducing frustration for both internal staff and any external systems integrating with the ERP platform.

BUSINESS REQUIREMENTS DOCUMENT

Product Schema

Prepared by Kolosal AI

Version 1.0 **Date** July 27, 2026 **Status** Draft **Classification** Internal

1. Data

1.1 Overview

This feature defines the complete "Product" record that the system uses whenever someone looks up, creates, edits, or changes the status of a product in the ERP. Rather than just storing basic product details, the system pulls together information from multiple related areas — pricing, stock levels across stores and warehouses, brand and category details, tags, and historical changes — so that anyone viewing a product gets a full, accurate picture in one place.

1.2 Step-by-Step Flow

1. Product List Request: When someone requests a list of products (for example, browsing a product catalog), the system retrieves matching products along with a total count, and returns them in manageable pages rather than all at once.
2. Single Product Lookup: When someone requests full details for one specific product at a specific store, the system gathers information from several sources at once: the store's stock on hand, the core product details, any tags attached to the product, warehouse inventory figures, the brand information, and the product type/category information.
3. Combining the Data: All of this gathered information is merged into one consolidated view, which is then cleaned up and reshaped into a single, easy-to-read Product record before being shown to the user.
4. Warehouse Stock Detail (Simple View): For a quick look at how much stock exists across warehouses for one product, the system can pull a simplified, ready-made stock summary.
5. Warehouse Stock Detail (Full Calculation): For a more detailed and accurate warehouse stock view, the system calculates real-time figures: how much inventory exists, how much is reserved for pending orders, how much is on its way in (in-transit), and how much is actually available to sell or move — along with cost information.
6. Product Change History: When someone wants to see how a product's pricing or status has changed over time, the system retrieves a chronological log showing what changed, when, and who made the change.

1.3 Key Concepts

- Product Record: The master set of information about a product — its name, brand, category, pricing, and status flags (such as whether it's active or price can be adjusted locally).
- Store Stock: How much of a product is currently available at a specific physical store.
- Warehouse Stock: How much of a product exists in a warehouse, further broken down into what's reserved, what's incoming, and what's actually available.
-

Reserved Quantity: Stock that has already been promised to an order and can't be sold again until that order is resolved.

- In-Transit Quantity: Stock that has been shipped but hasn't yet arrived at its destination warehouse.
- Tags: Optional labels attached to a product to help with searching, filtering, or grouping.
- Brand & Type: Reference information that categorizes a product by manufacturer (brand) and product category (type).
- Product History: A running log that records every meaningful change made to a product's price or status, including who made the change and when — useful for audits and troubleshooting.
- Pagination: A way of splitting large result sets (like long product lists) into smaller, manageable pages instead of loading everything at once.

1.4 Business Value

This feature ensures that everyone working with product data — whether in a store, warehouse, or head office — sees one consistent, trustworthy version of the truth about each product. By combining pricing, stock, and categorization data into a single view, it reduces the risk of staff working with outdated or incomplete information, speeds up decision-making (such as whether to restock or promote a product), and provides an audit trail for accountability when prices or statuses change. It also supports scalability, since large product catalogs and multi-warehouse operations can be handled efficiently through pagination and real-time stock calculations.

BUSINESS REQUIREMENTS DOCUMENT

Common Standard Response JSON Schema

Prepared by Kolosal AI

Version 1.0	**Date** July 27, 2026	**Status** Draft	**Classification** Internal

1. Data

1.1 Overview

This feature defines a shared "rulebook" for how the ERP system's software components respond whenever an action is performed through the system's programming interface (API). Instead of every part of the system inventing its own way to report success or failure, this feature establishes one consistent, standardized format for every possible outcome — whether something worked, was rejected, or failed unexpectedly. This ensures all parts of the ERP system "speak the same language" when communicating results.

1.2 Step-by-Step Flow

1. Start with a master reference list of all the standard outcome types the system can return (for example: "Created," "Not Found," "Conflict," "Too Many Requests," etc.), each with a code number, a name, and a plain description of when it applies. This list acts as a glossary that all other formats connect back to.
2. Define the format used when something is successfully created — it includes the outcome code, the new information that was created, and, optionally, a reference link to find that new item later.
3. Define the format used when a request is invalid or poorly formed — it includes the outcome code, a summary message, and optionally a list of specific problems found.
4. Define the format used when someone tries to access something without proper login credentials — it includes the outcome code and an explanation message.
5. Define the format used when someone is logged in but not allowed to perform the action — it includes the outcome code and an explanation message.
6. Define the format used when the requested item cannot be found — it includes the outcome code and a message identifying what was missing.
7. Define the format used when an action conflicts with the current state of something (for example, trying to create a duplicate) — it includes the outcome code, a fixed label, a detailed message, and optional extra context.
8. Define the format used when a request looks correct but fails business rule checks — it includes the outcome code, a short label, a summary message, and an optional detailed list of which specific fields had problems and why.
9. Define the format used when someone has made too many requests too quickly — it includes the outcome code, a message, and optionally how long they should wait before trying again.
10. Define the format used when the system encounters an unexpected internal problem — it includes the outcome code and a general error message.
11.

Define the format used when the system is temporarily unavailable (for example, during maintenance or overload) — it includes the outcome code and an explanation message.

12. Every one of these formats links back to the master reference list from step 1, ensuring the code, name, and meaning stay consistent everywhere they're used.

1.3 Key Concepts

- Response envelope: A consistent, predictable "container" format used to wrap every answer the system sends back, no matter the outcome.
- Status code: A standardized number (like 201, 404, 500) that quickly signals what kind of outcome occurred — success, client error, or system error.
- Required vs. optional fields: Some pieces of information must always be included in a response (like the outcome code and a message), while others are only included when relevant (like a retry wait time).
- Reference lookup: A central glossary of all outcome types, ensuring every team and system component uses the same definitions and terminology.
- Validation error details: Extra structured information explaining exactly which piece of submitted data was problematic and why, useful for pinpointing input mistakes.

1.4 Business Value

By standardizing exactly how the system reports success and failure, this feature reduces confusion, miscommunication, and costly integration errors between different teams and systems (such as the technical teams building the backend logic and those building the user-facing screens). It ensures a predictable, professional experience for anyone or any system interacting with the ERP platform — users see clear, consistent explanations when something goes wrong, and developers can build and troubleshoot faster because every response follows the same trusted structure. This consistency ultimately lowers development costs, speeds up delivery, and improves the reliability of the overall ERP system.

BUSINESS REQUIREMENTS DOCUMENT

Product

Prepared by Kolosal AI

Version 1.0 **Date** July 27, 2026 **Status** Draft **Classification** Internal

1. Data

1.1 Overview

This feature defines how product information is organized and stored within the ERP system. It sets up the "Product" as the central piece of information that everything else in the system connects to — including what brand and type a product belongs to, how it's priced in different locations, how its stock is tracked across stores and warehouses, and a complete history of changes made to it over time. In short, this is the foundation that lets the business reliably track every product it sells, from creation through pricing changes to stock movements.

1.2 Step-by-Step Flow

1. Set up lookup lists first: Before a product can be created, the system needs two reference lists — Product Brands (e.g., Toyota, Honda) and Product Types (e.g., Spare Part, Accessory). These act like dropdown menus that keep naming consistent across all products.
2. Create the core Product record: Each product is registered with essential details — a unique SKU code, product name, which brand and type it belongs to, whether it's currently active for sale, whether stores can override its price, its tax category, and whether its stock is automatically tracked.
3. Track every price and status change: Whenever a product's pricing (supply price, wholesale price, retail price, trade-in value, warranty) or its active/tracked/bundle status changes, the system automatically logs a snapshot of that change, including who made it and when — creating a permanent, unchangeable history trail.
4. Apply location- or group-specific pricing: Products can have different override prices depending on where they're sold or who's buying — by region, specific store, specific warehouse, or customer group — allowing flexible pricing strategies without altering the base product record.
5. Tag products for easy grouping: Products can be labeled with tags (e.g., "On Sale," "New Arrival") to help with searching, filtering, and marketing efforts.
6. Record every purchase transaction: Every time stock is purchased from a supplier, a permanent record is kept — including quantity, price, tax, and total cost — creating a full purchasing history per product.
7. Track stock at the warehouse level: Each warehouse keeps a running count of how much of a product it has in inventory, how much is reserved for pending orders, how much is on consignment, and how much is actually available to sell.
8. Track stock at the store level: Similarly, each store keeps its own running stock count for every product.
9. Log every store stock change: Any time stock at a store increases or decreases (due to a sale, stock transfer, or other event), the system records the event details — what changed, by how much, and why — before updating the store's current stock total.

1.3 Key Concepts

- Product: The central record representing an item the business sells, holding its identity, pricing rules, and status flags.
- SKU: A unique code used to identify and look up a product across all parts of the system — like a product's fingerprint.
- Brand & Type: Standardized categories that describe what a product is and who makes it, kept in separate reference lists to avoid inconsistent naming.
- Price Tracking: A way to set different prices for the same product depending on the region, store, warehouse, or customer group it's sold to.
- Product History: A permanent audit trail showing how a product's price and status have changed over time, along with who made each change.
- Purchase History: A permanent log of every time stock of a product was bought from a supplier, including cost and tax details.
- Warehouse Stock / Store Stock: Running totals showing how much of a product is available at each warehouse or store.
- Store Stock History: A permanent log of every stock adjustment at a store, showing what caused the change (a sale, transfer, etc.).
- Product Tags: Labels attached to products to support grouping, searching, and marketing.
- Frozen Product: A status that prevents a product from being moved between branches, typically used to control inventory movement temporarily.

1.4 Business Value

This structure gives the business a single, reliable source of truth for all product-related information, reducing errors caused by duplicate or inconsistent data. Automatic history tracking for pricing and stock changes ensures full accountability and makes it easy to answer "who changed what, and when" — valuable for audits, dispute resolution, and financial reporting. Flexible pricing by region, store, or customer group enables smarter sales strategies without complicating the core product setup. Meanwhile, real-time stock visibility across both warehouses and stores helps prevent overselling, supports better purchasing decisions, and improves overall inventory accuracy — all of which contribute to smoother operations and better customer service.